

5

RESISTANCE AND RESISTORS

This chapter explains the factors that determine the resistance of a conductor. It also introduces the resistor, a commonly used component in electrotechnology, and describes a number of other components whose resistance varies with an external influence, such as light or heat. The resistor colour code is explained, along with how to measure resistance with an ohmmeter.

Chapter outline

5.1 Factors that determine resistance

5.2 Resistors

5.3 Resistor colour code

5.4 Measuring resistance



5.1 Factors that determine resistance

Electrical resistance is a property of all conducting materials, and in the case of a cable, resistance should be as low as possible. Resistance is sometimes required, such as in a heating element and also in components called resistors. We look first at conductors and the factors that determine their resistance. The four factors are:

- material the conductor is made of
- its length
- its cross-sectional area
- the temperature of the conductor.

Material the conductor is made of

The term resistance first appeared in Chapter 1 where it was explained that certain materials are better conductors than others, due to their atomic structure. This is a property of the particular material, and is referred to its *resistivity*. All conducting materials have a certain value of resistivity, which is defined as the resistance the material offers between opposite faces of a cube of the material, as shown in Figure 5.1. In the metric system, the cube has one metre dimensions and the resistivity value is expressed in *ohm-metres* (Ωm). The symbol for resistivity is the Greek letter ρ (pronounced rho).

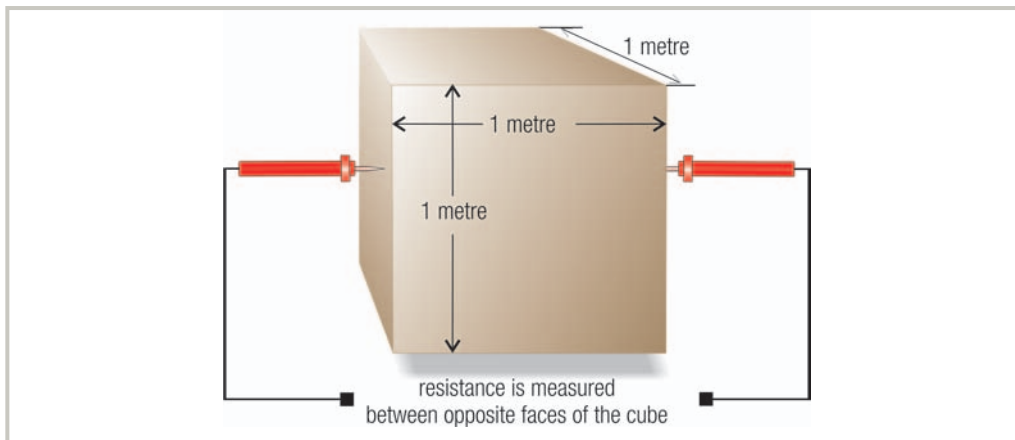


FIGURE 5.1

Resistivity of a material is its resistance measured at a specified temperature between opposite faces of a cube of the material

In practice, resistivity is measured with a smaller sample of the material, with the value expressed in terms of a one metre cube. Resistivity, when given for a particular temperature, gives us a value that is independent of all other factors that determine the resistance of a conductor. Table 5.1 lists resistivity values for typical conductors used in electrotechnology. These values are for a temperature of 20 °C, because resistivity varies with temperature. The purity of the material also affects the resistivity value, and you may find tables in other books give slightly different values.

A cube with one metre dimensions is quite large, so you get very small resistivity values. These are, therefore, expressed with scientific notation. The figure for silver is the lowest (as it's the best conductor) and carbon has the highest resistivity. Carbon is not a conductor, and is found in some types of resistors. It is included here for comparison. In fact, carbon has a resistivity some 200 000 times higher than silver. Nichrome is only about 69 times greater than silver, but this is high enough to make its resistance suit a heating element.

TABLE 5.1

RESISTIVITY OF CONDUCTORS AT 20 °C		
Material	Resistivity in ohm-metre	
silver	1.63×10^{-8}	Note: Expressed without scientific notation, silver has a resistivity of 0.0000000163 ohm-metre
copper	1.72×10^{-8}	
gold	2.44×10^{-8}	
aluminium	2.83×10^{-8}	
tungsten	5.3×10^{-8}	
nichrome	112×10^{-8}	
carbon	3.5×10^{-3}	

Table 5.2 shows the approximate resistance of a 1 km (1000 m) length of 1 mm diameter wire made from each of the materials in Table 5.1. As explained in Chapter 2, only pure copper produced by electrolysis is found in electrical conductors. Aluminium conductors are used in high voltage power transmission lines. Tungsten is often the filament material in an incandescent lamp and nichrome is used in electric radiator elements.

TABLE 5.2

COMPARISON OF CONDUCTOR RESISTANCE	
Material	Resistance in ohms (1 km by 1 mm dia. wire at 20 °C)
silver	20.4
copper	21.7
gold	28.0
aluminium	34.4
tungsten	67.5
nichrome	1426
carbon	4 456 327

Resistance is proportional to the resistivity of the material a conductor is made of



Resistance is proportional to the length of the conductor



Length of the conductor

Electrical resistance is associated with how easily electrons can flow from one atom to another within the material. The easier this is, the lower the energy losses each time an electron moving in a conductor encounters an atom. As the length of a conductor increases, its resistance to the path of electrons becomes greater, as there are now more electron-atom collisions, and therefore increased losses.

In Table 5.2, if the conductor length is doubled, the resistance values would also double. That is, from the table, a 2 km length of 1 mm diameter copper wire would have a resistance of 43.4 Ω . If its length is halved to 500 m, the resistance will also halve, giving 10.85 Ω .

Cross-sectional area

Conductors can be any shape including circular (wire), rectangular or square. While diameter is a useful measurement for wire, it doesn't apply to other shapes. However, all conductors have a cross-sectional area (csa), regardless of their shape. The three conductors in Figure 5.2 all have the same cross-sectional area of 4 mm^2 .

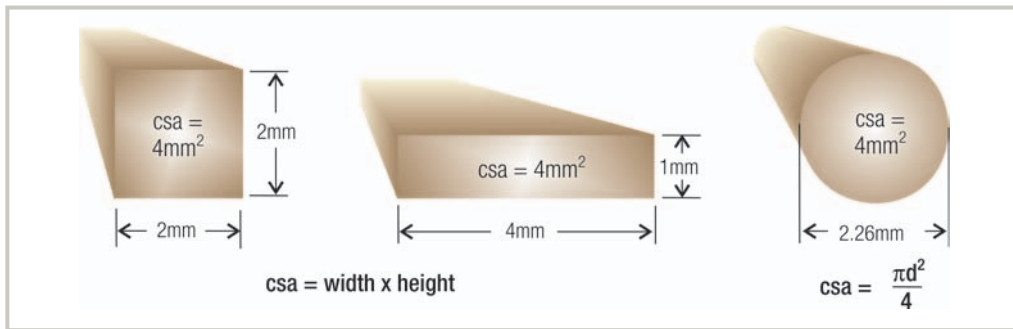


FIGURE 5.2

Cross-sectional area of a conductor is the area of the face of the conductor

Notice how the cross-sectional area of each of the conductors is determined. For a rectangular or square conductor, the csa is the height multiplied by the width. For a circular conductor $\text{csa} = \frac{\pi d^2}{4}$, which, as you probably recognise, is the equation to find the area of a circle.

The smaller the cross-sectional area the higher the resistance, as there are less free electrons to carry the current. If the cross-sectional area is increased, the resistance decreases. Therefore, the resistance of a conductor is *inversely* proportional to its cross-sectional area.



Resistance is inversely proportional to cross-sectional area

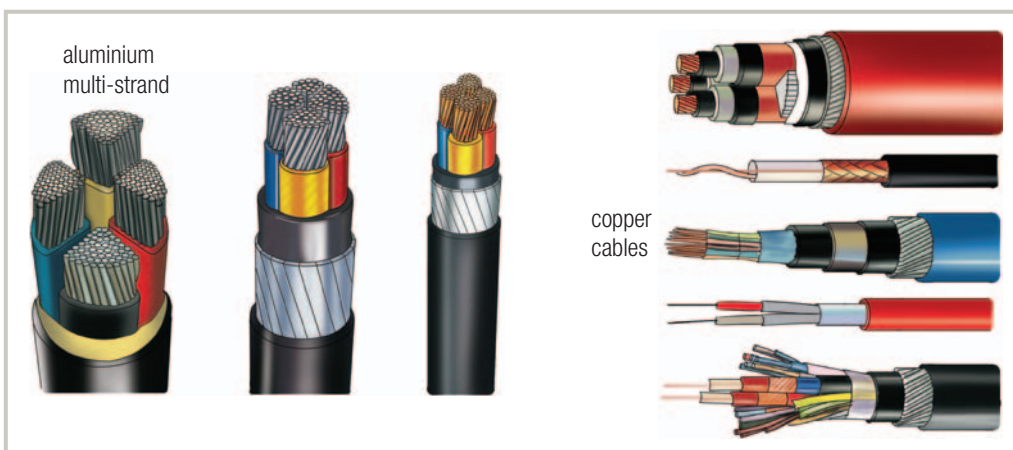


FIGURE 5.3

Electrical cables come in a wide variety of sizes, shapes and types. Cables are usually multi-stranded and made of copper or (less typically) made of aluminium.

Calculating cable resistance

Length, cross-sectional area and resistivity can be combined into an equation to calculate the resistance of a conductor. The equation is:

$$R = \frac{\rho l}{A}$$

where:

R = resistance in ohms

ρ = resistivity in ohm-metres

l = length in metres

A = cross-sectional area in square metres.

In most cases, the cross-sectional area of a conductor is given in square millimetres (mm^2). Because there are 1000 or 10^3 millimetres in a metre, a square metre (m^2) can be expressed as 10^6 square millimetres. Therefore, to convert from square millimetres to square metres, multiply the value given in square millimetres by 10^{-6} .

EXAMPLE 5.1

What is the resistance of a 1 km length of copper cable with a csa of 2 square millimetres (mm^2)?

Solution

Values: From Table 5.1, resistivity (ρ) of copper = 1.72×10^{-8} ohm-metre
 csa = $2 \text{ mm}^2 = 2 \times 10^{-6}$ square metres (m^2)
 length (l) = 1 km = 1000 m

$$\text{Equation: } R = \frac{\rho l}{A} = 1.72 \times 10^{-8} \times \frac{1000}{2 \times 10^{-6}}$$

Answer: **$R = 8.6 \Omega$**

EXAMPLE 5.2

Calculate the resistance of a nichrome resistance element made from 1 mm diameter wire. The total wire length is two metres.

Solution

Values: From Table 5.1, resistivity (ρ) of nichrome = 112×10^{-8} ohm-metre
 length (l) = 2 m
 diameter = 1 mm

$$\text{Equation 1: } A = \frac{\pi d^2}{4} = \frac{3.14 \times 0.001^2}{4} = \frac{3.14 \times 10^{-6}}{4} = 7.85 \times 10^{-7} \text{ sq metres}$$

$$\text{Equation 2: } R = \frac{\rho l}{A} = 112 \times 10^{-8} \times \frac{2}{7.85 \times 10^{-7}}$$

Answer: **$R = 2.85 \Omega$**

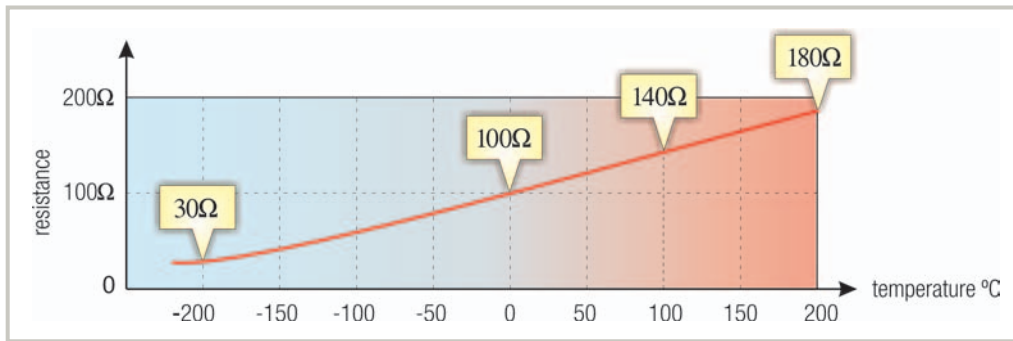
Temperature

The resistance of all conductors varies with temperature. For pure metals, an *increase* in temperature will give an *increase* in resistance. How much the resistance changes depends on the type of metal.

The change in resistance can be calculated with a value called the **temperature coefficient of resistance** for that metal. Temperature coefficient of resistance is defined as the change in resistance per ohm per degree Celsius of temperature change.

The graph in Figure 5.4 shows how the resistance of a coil of copper wire changes with temperature. If the resistance at 0°C is 100Ω , it will be 180Ω at 200°C . However, if the temperature drops to -200°C , the resistance will drop to about 30Ω .

The graph is a straight line for temperatures above 0°C . Because the resistance increases with an increase in temperature, the temperature coefficient is *positive*. Most metals have a positive temperature coefficient of resistance. If the temperature is reduced to absolute zero (-273°C), the resistance of many types of metal conductors falls to virtually zero. This is known as *superconductivity*.

**FIGURE 5.4**

Pure metals increase their resistance with an increase in temperature

Some materials, such as carbon and silicon (semiconductors), have a negative temperature coefficient of resistance. This means their resistance *drops* as the temperature rises.

The temperature coefficient for a material is given at a particular temperature, such as 20 °C. This is necessary as the resistance of the material changes with temperature, and the temperature coefficient of resistance drops as the temperature rises. Otherwise, the graph above wouldn't be a straight line.

For most purposes, the value given at 20 °C can be used and for copper, aluminium, gold and silver the value is about 0.004. This means the resistance will increase by about 0.4 per cent for an increase of 1 °C. In Figure 5.4, the resistance of the coil has increased by 40 per cent from 100 Ω to 140 Ω for a temperature rise of 100 °C.

Table 5.3 lists the temperature coefficient of resistance for a range of conductors. The purity and the composition of the material will affect these values, and you may find that similar tables in other books show different values.

TEMPERATURE COEFFICIENTS	
Material	Temperature coefficient per degree C at 20 °C (α_{20})
silver	0.004
copper	0.00392
gold	0.0034
aluminium	0.0039
tungsten	0.0045
nichrome	0.0017
carbon	−0.0005

TABLE 5.3

Carbon has various forms and, depending on its composition, temperature coefficients for carbon will vary considerably. Most types of carbon have a *negative* temperature coefficient (resistance falls as temperature rises), although some types have a positive temperature coefficient. The value given is for amorphous carbon. Notice that the value is negative.

The symbol for temperature coefficient is the Greek letter α (alpha). Because α varies with temperature, the reference temperature is usually shown as a subscript, for example α_{20} means the temperature coefficient for the material at a temperature of 20 °C.

The effect of temperature can sometimes be significant. For example, the resistance of a typical 40 W, 230 V incandescent lamp is about 100 Ω when the lamp is turned off. This is the lamp's *cold* resistance. It increases to 1440 Ω when the lamp is operating. Therefore, the current taken by the lamp is much higher when power is first applied, compared to the normal operating current when the lamp has reached its operating temperature.



For pure metals, resistance is proportional to temperature

With spot welding, when a large current is passed through the point being welded, the resistance of the metal generates heat. Aluminium has a low resistance, so it's harder to spot weld than stainless steel, which has a higher resistance and therefore heats up quickly.

SUMMING UP...

- Resistance is proportional to the length (in metres) of the conductor.
- Resistance is proportional to the resistivity of the material a conductor is made of.
- Resistivity (ρ) is measured in ohm-metres.
- Resistance is inversely proportional to cross-sectional area (csa) of a conductor.
- Cross-sectional area is in square metres (m^2).
- For pure metals, resistance is proportional to temperature.
- Copper and aluminium have a positive temperature coefficient of 0.004.

EXERCISE 5.1

- 1 Find the resistance of a 50 metre length of copper cable with a cross-sectional area of 2.5 mm^2 .
- 2 If the length of the cable in Question 1 is increased to 500 metres, what is the resistance of the cable?
- 3 An 800 metre long aluminium cable has a diameter of 3 mm. Calculate its resistance.
- 4 If the cable in Question 3 is made of copper, what is its resistance?
- 5 Find the resistance of a two metre length of 0.5 mm diameter tungsten wire.

5.2 Resistors

A resistor is an electrical component with a fixed value of resistance. Resistor values range from less than one ohm up to many millions of ohms. Resistors are also given a *power rating* because they get hot when current is passed through them. The power rating of a resistor determines the maximum current the resistor can pass without overheating.

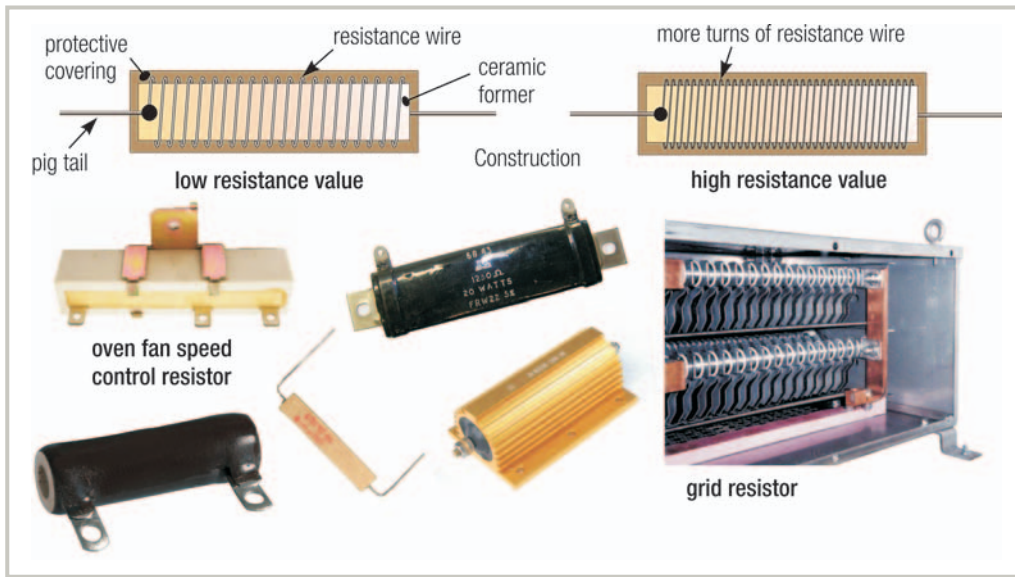
High power resistors

Resistors with a rating of 5 W or more are generally made by winding resistance wire around a ceramic former, rather like a radiator element. These resistors are known as wire-wound types. Some are fitted with heatsinks to dissipate the heat generated by the resistor, and these must be mounted so heat can be dispersed easily.

Another type is the *grid resistor*, which is designed to handle high currents (400 A or more) and high power dissipation. These have large resistive elements made from a metal alloy, and are used in applications such as dynamic braking and speed control of large motors, load banks for testing purposes and in electrical substations. Power resistors are shown in Figure 5.5.

Film resistors

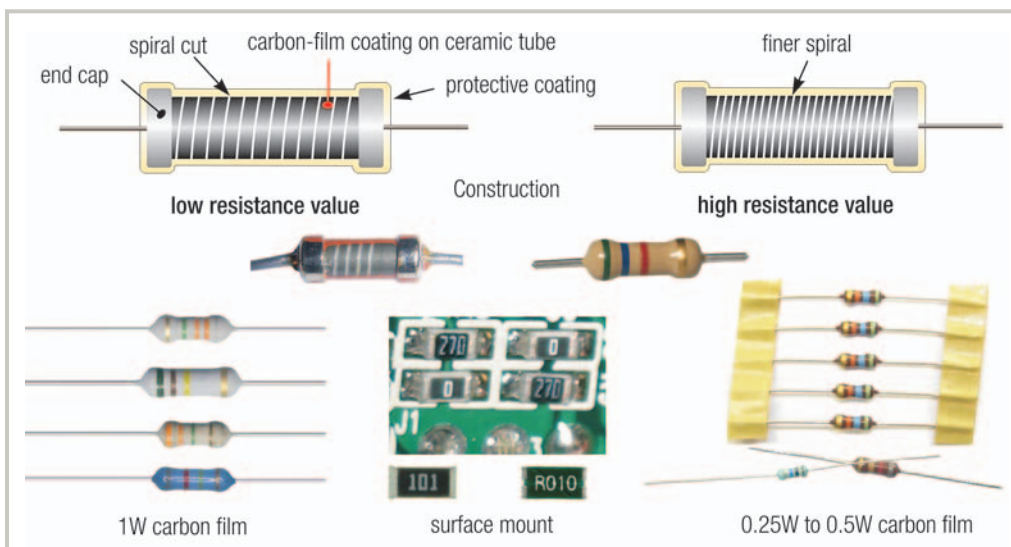
Low power resistors are made by coating a ceramic cylinder with a film of specially prepared carbon. A spiral is then cut in the film to give a longer path. These resistors are called *carbon-film* resistors and the higher the value, the finer the spiral for a given type of coating. These resistors are used extensively in electronic applications, although they are often found in electrical work. They are the most common type of general purpose resistor.

**FIGURE 5.5**

Power resistors, showing typical construction and types of resistors

A variation is the *metal-film* resistor, which has a similar construction to the carbon-film type, except the resistive element is a coating of nickel chromium or some form of metal oxide. These resistors have a higher stability than carbon-film types, and are made using thin-film technology in which the coating is deposited by a vacuum process. Surface mount resistors are made by depositing a metal oxide layer onto a rectangular ceramic substrate, which has contacts at either end. Surface mount resistors (SMD) are widely used in mobile phones, computers and most electronic devices.

In general, the size of a film resistor gives a guide as to its power rating. Typically, these sorts of resistors have power ratings up to 5 W, the most common being $\frac{1}{4}$ W, $\frac{1}{2}$ W and 1 W. Figure 5.6 shows the construction of a carbon-film resistor, along with typical examples.

**FIGURE 5.6**

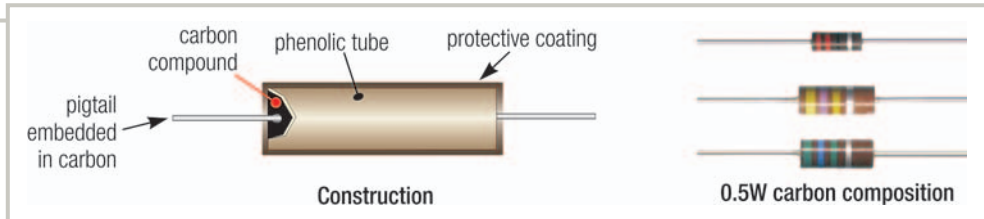
Carbon-film resistors are for low power use. They range in power from $\frac{1}{4}$ W to 5 W, and cost a few cents each.

Carbon composition resistors

An older style of resistor is the *carbon composition* type, in which powdered carbon mixed with a filler is compressed inside a phenolic cylinder, with connections (pigtails) at each end. These resistors are used in applications involving high-energy current pulses, as they can withstand this type of stress better than the carbon-film types. However, they tend to change their resistance value over time. Their construction is shown in Figure 5.7.

FIGURE 5.7

Carbon composition resistors were the first types of resistors to be made. They have limited application today, but are still manufactured.



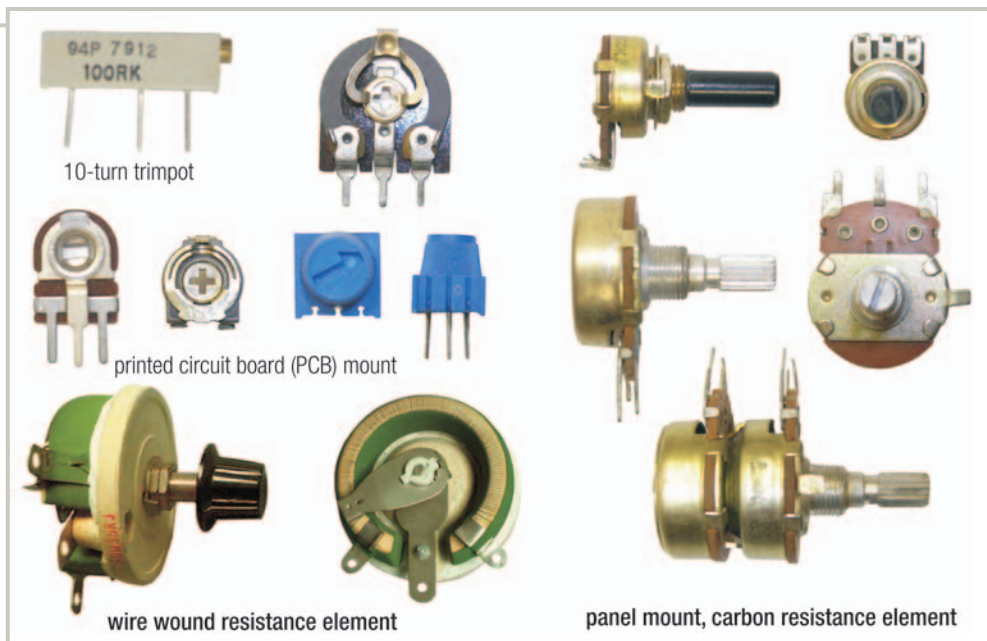
Variable resistors

A *variable resistor* is used in a circuit when you need to be able to adjust the value of a current or a voltage. Variable resistors are often called potentiometers ('pots' for short) because they are generally used to adjust a voltage, or potential difference, in a circuit. This requires them to have three terminals: one at either end of the resistance element, and another that connects to the moving contact, or wiper.

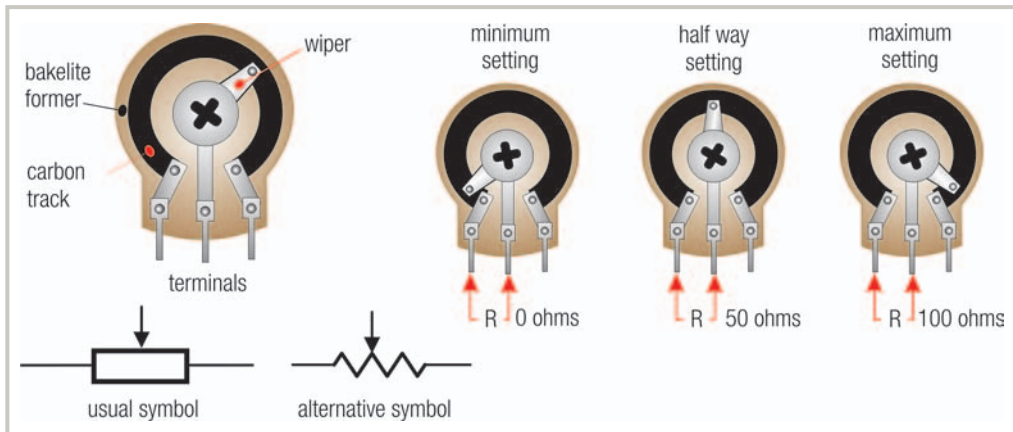
Variable resistors come in a wide range of sizes, resistance values and power ratings. They have a resistance element that is either wire-wound, or made of a carbon or metal compound deposited onto a substrate. As in a wire-wound resistor, the resistance element in a wire-wound variable resistor is made from resistance wire wound around a heat resistant ceramic former, allowing it to handle higher values of current and dissipate more power. Most variable resistors have a carbon track resistance element. As shown in Figure 5.8, there are two main types: panel mount and printed circuit board (PCB) mount. The dual-gang panel mount pot has two variable resistors on the one shaft, and is used in stereo (two channel) sound systems. Those used on a printed circuit board are often called 'trim pots', as they are adjusted with a screwdriver.

FIGURE 5.8

Typical PCB and panel mount potentiometers



The basic construction of a carbon-track variable resistor is shown in Figure 5.9. The resistance element is made from a carbon compound deposited onto a bakelite former. A wiper contacts the carbon track. The wiper is mechanically coupled to the shaft and is also electrically connected to the centre terminal. The ends of the carbon track are connected to the outer terminals, and its resistance value is printed or stamped on the case.

**FIGURE 5.9**

A variable resistor (potentiometer) has three terminals, the centre one connected to a moving wiper. As the wiper is moved along the carbon track, the resistance between it and the outside terminals changes.

As Figure 5.9 shows, a variable resistance value is obtained by moving the wiper from one side of the resistance element to the other. If the carbon track has a *linear* resistance, when the wiper is halfway, there will be equal resistance between the centre terminal and the outside terminals.

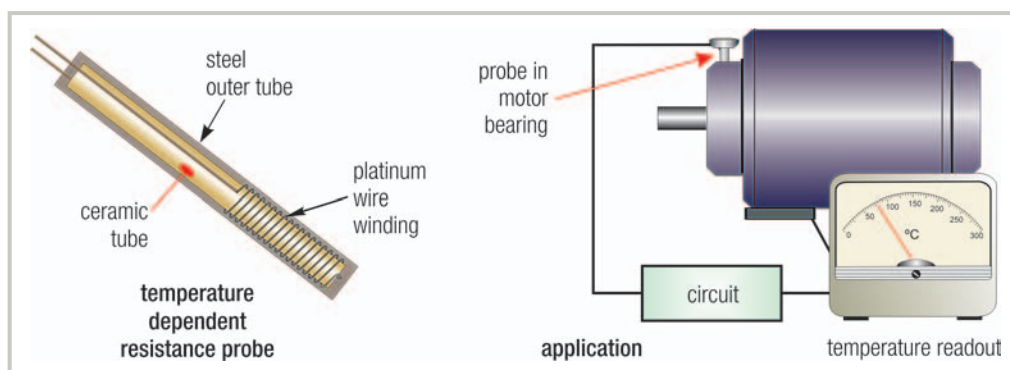
However, in some types of potentiometers, the change in resistance is *non-linear*. That is, when the wiper is halfway, there is a considerable difference in resistance between it and the outside terminals. The most common use for a non-linear variable resistor is the volume control in a radio. Because human hearing is *logarithmic*, the resistance element in a volume control is constructed so its resistance changes in a logarithmic way, to make the change in sound level appear to follow the change in the position of the control.

The Australian Standard symbol for the variable resistor is shown in Figure 5.9. This and the symbol for a fixed resistor were introduced in Chapter 3. As with the resistor, the rectangular symbol for a variable resistor has replaced the older 'wiggly line' symbol, although some overseas equipment manufacturers use this symbol.

Temperature-dependent resistors

We've shown that temperature affects the resistance of a conducting material. Normally this is not wanted, but use can be made of the effect. Resistors that change their resistance value with a change in temperature have a number of applications. For example, they are often used as the sensor in a temperature measuring system, an over-temperature protection system or a temperature control system.

A temperature-dependent resistor can be made by winding copper, nickel or platinum wire around a ceramic former as shown in Figure 5.10. The resistive element is placed inside a protective sheath, usually made of metal. The whole assembly can then be used as a probe in a temperature measuring, protection or control system. The change in resistance for each degree of temperature change is fairly small, although it will be linear. For example, if the resistance is $110 \, \Omega$ at $20 \, ^\circ\text{C}$, it will be about $140 \, \Omega$ at $100 \, ^\circ\text{C}$.

**FIGURE 5.10**

A temperature probe with a resistance winding of thin platinum wire

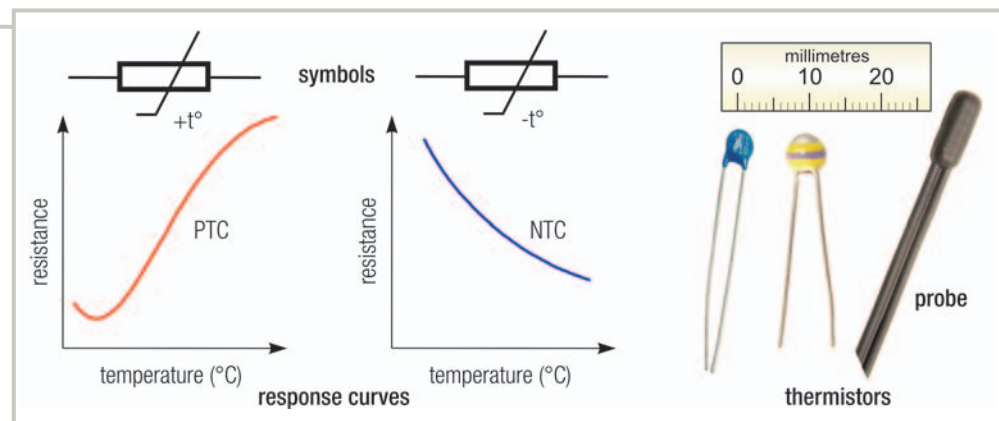
Another very commonly used temperature dependent resistor is the *thermistor* (thermal resistor). These are made with a semiconductor material and have either a positive temperature coefficient (PTC) or a negative temperature coefficient (NTC). The first NTC thermistor was discovered in 1833 by Michael Faraday, who reported the effect temperature had on the resistance of silver sulphide. However, commercial production of thermistors did not begin until the 1930s.

They give a large change in resistance over their operating range, although they're limited to where the temperature doesn't exceed a few hundred degrees Celsius. Thermistors are generally packaged in a protective coating, with two connecting leads. They are physically small, and are used in some types of temperature probes.

Figure 5.11 shows some typical thermistors, how their resistance changes with temperature and the symbols for NTC and PTC types. The curve for the PTC type shows that the resistance actually drops for a small rise in temperature, then rises rapidly once a certain temperature is reached. As the curves show, thermistors have a linear response only for small temperature changes.

FIGURE 5.11

Thermistors have many applications, and are either NTC or PTC devices packaged in disc or rectangular form



Thermistors are often used to protect an electric motor against overheating. In small electric motors, a PTC thermistor is embedded inside the windings and connected so the motor current passes through the thermistor. If the temperature of the windings rises above a certain value, the resistance of the thermistor will quickly increase (from a low resistance to several thousand ohms), reducing the motor current to almost zero.

In other words, the thermistor behaves like a switch that turns off when the temperature is too high. When the motor cools down, the resistance of the thermistor drops, letting the motor run. In larger motors, the thermistor is connected to a switch that cuts off power to the motor.

Light-dependent resistors

Light-sensitive devices have a number of uses in electrotechnology. The light-dependent resistor (LDR) consists of a thin ceramic disc sintered with cadmium-sulphide. (Sintering is the process of using heat to combine two materials.) Cadmium-sulphide is a *photo-conductive* material, which means its resistance is affected by light. A vacuum-deposited metallic grid is applied to the surface of the disc and the whole assembly then covered in clear plastic.

The resistance of LDRs in complete darkness is over 10 million ohms (10 MΩ), which falls to less than 100 Ω in normal daylight. They are often used as a light sensor in automatically controlled lighting installations, where the increase in resistance of the LDR causes the lights to switch on at sunset. The LDR is connected to operate a relay, which in turn switches the lights. When it's daylight, the resistance of the LDR falls, switching off the relay that controls the lights.

The construction, symbol and response curve of an LDR are shown in Figure 5.12. A typical LDR is about the size of a five cent piece. As the response curve shows, the LDR is a non-linear resistor.

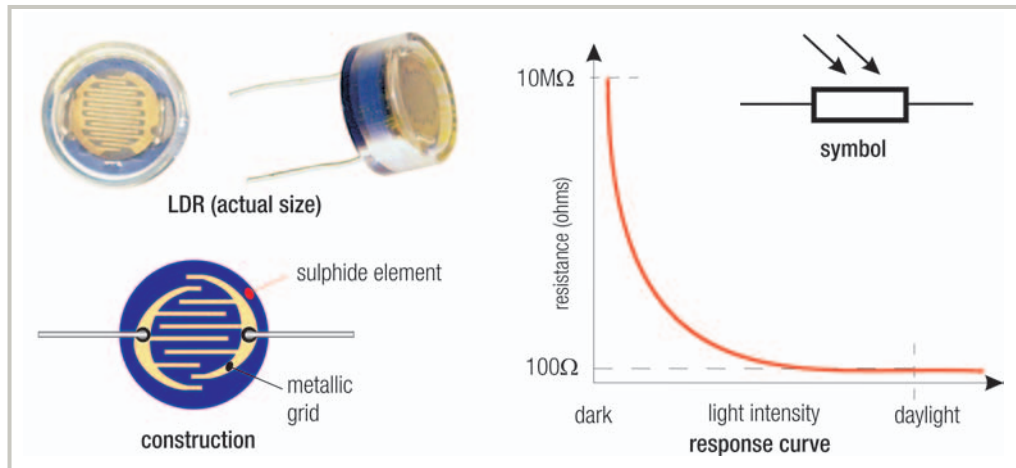


FIGURE 5.12

The LDR symbol, construction and response curve

Voltage-dependent resistors

A voltage-dependent resistor (VDR or *varistor*) is a component in which the voltage across the device significantly affects its resistance. At its rated breakdown voltage, a varistor becomes a very low resistance. When the voltage is less than the breakdown voltage, the device is virtually an open circuit. It therefore behaves more like a switch than a resistor. They are often known as surge protectors.

Most varistors are shaped as discs or rods and are made from zinc oxide mixed with small amounts of other metal oxides. Varistors made this way are called metal oxide varistors or MOVs. Varistors are also made from a sintered compact of silicon carbide particles. Conducting leads are attached to the sides of a disc or the ends of a rod, and the assembly is fired at high temperature, then covered with protective insulation. Figure 5.13 shows typical disc MOV VDRs that are used as surge protectors in electronic equipment.

The main use of VDRs is to protect a circuit from a voltage surge. When the voltage across a VDR goes over a certain value, its resistance drops within milliseconds to a very low value. If the voltage surge lasts long enough, the large current flowing in the VDR will blow the circuit fuse (or trip a circuit breaker) and isolate the circuit from the supply. If the surge is very brief, the energy in the surge will be dissipated by the VDR without blowing the fuse.

A VDR is therefore given two ratings: the breakdown voltage and the energy (in joules) it can dissipate. Voltage ratings vary from 5 V to several hundred volts. A typical voltage rating for a VDR connected across the 230 V AC mains supply is 275 V. A VDR is connected across (or in parallel with) the circuit it is protecting, as illustrated in Figure 5.14. Notice that the VDR is connected *after* the fuse.

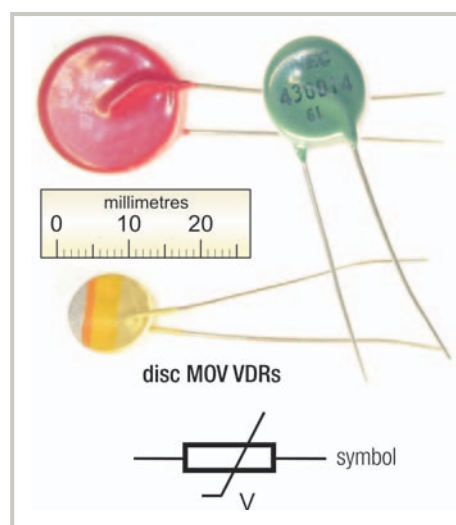


FIGURE 5.13

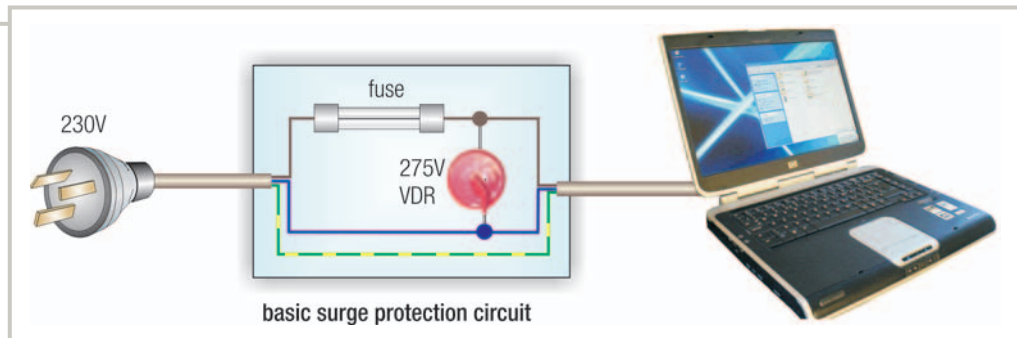
Typical VDRs (varistors) used as voltage surge protectors



Surge protector plugs are fitted with a VDR

FIGURE 5.14

A VDR is connected in parallel and the rise in voltage allows a fault current to flow which ruptures the fuse

**FIGURE 5.15**

High voltage power line surge arrester



Surge arrestors

In electrical power generation, transmission and distribution, equipment is protected against lightning strikes by devices called lightning or *surge arrestors*. These are similar in construction to the VDRs described above, except they are much larger and can absorb high amounts of energy. They are connected between power lines and a metal stake driven into the ground. If the lines are struck by lightning, the arrestors divert the voltage surge to the ground, minimising or preventing damage to equipment. The response time of a surge arrester is very fast – important if it's to give enough protection from the voltage surge caused by a lightning strike.

SUMMING UP...

- Resistors are rated by their resistance and power handling capability.
- High power resistors have either a wire-wound or a metal grid resistance element.
- Low power (up to 5 W) resistors typically have a carbon-film resistance element.
- A variable resistor (also called a potentiometer, or pot) has either a wire-wound or carbon track resistance element, which can be linear or non-linear.
- A temperature dependent resistor changes resistance with temperature.
- If the resistance *increases* with a temperature increase, the device has a positive temperature coefficient (PTC). If it *decreases*, it has a negative temperature coefficient (NTC).
- Temperature sensors using this effect are either the resistive element wire-wound type (linear PTC), or a thermistor (NTC or PTC).
- A light dependent resistor has a cadmium sulphide element, which in full darkness has a high resistance. In daylight the resistance falls to around 100 ohms.
- A varistor (VDR) changes from a high resistance to a very low resistance when its breakdown voltage is exceeded. When the voltage is removed, the device returns to being a high resistance.
- Small metal oxide varistors (MOV) are used as surge protectors in electronic equipment. They are rated by breakdown voltage (5 V to over 500 V) and by their ability to dissipate short bursts of energy.
- Large metal oxide varistors are used as surge arrestors in the electrical power industry.

5.3 Resistor colour code

The value of a resistor is either printed on its body as a numerical value, or as a code of coloured bands. Each colour represents a number and the resistance value is determined by 'reading' the colour code. There are two types of resistors that use colour codes: general purpose resistors and precision resistors. The difference between these two resistors is their *tolerance* value.

Tolerance refers to how close the actual value of the resistor is to its marked value. For example, a resistor might be coded as having a resistance of 1000 ohms with a tolerance of ± 10 per cent. This means the maker guarantees the actual value won't be more than 10 per cent higher or lower than 1000 ohms. As 10 per cent of 1000 is 100, the actual value of the resistor could be anywhere between 900 ohms ($1000 - 100$) to 1100 ohms ($1000 + 100$).

General purpose resistors are those with a tolerance of ± 5 per cent or more. Precision resistors have a tolerance of less than 5 per cent, usually ± 2 per cent or ± 1 per cent, even as low as ± 0.05 per cent. General purpose resistors show their value with four bands, and precision resistors have five bands. The last band is always the tolerance band, and is spaced further apart from the other bands. Figure 5.16 shows these two types of resistors and what each band represents. There are no standards concerning the body colour of a resistor, although some manufacturers use body colour to identify power rating.

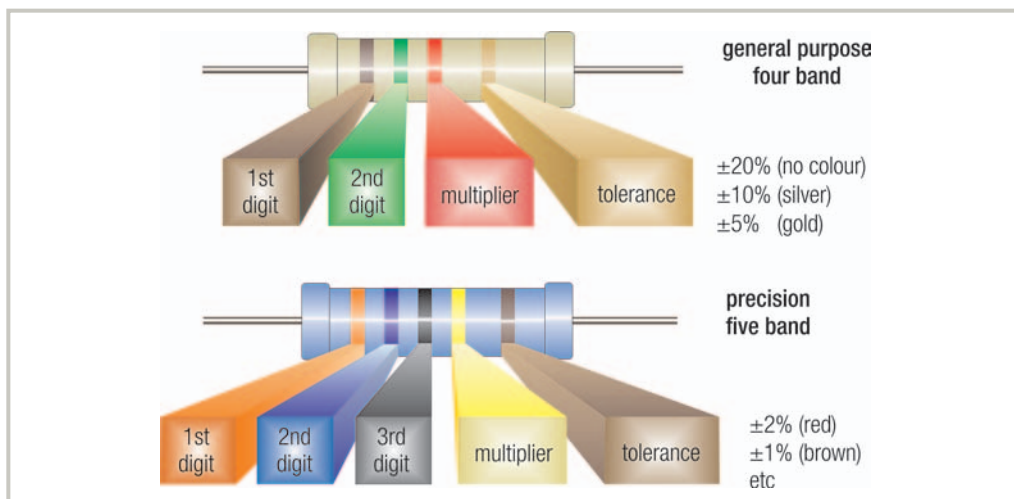


FIGURE 5.16

Colour bands for four and five band resistors

Figure 5.17 shows the 12 colours used in the resistor colour code. Gold and silver are only used to show a tolerance value.

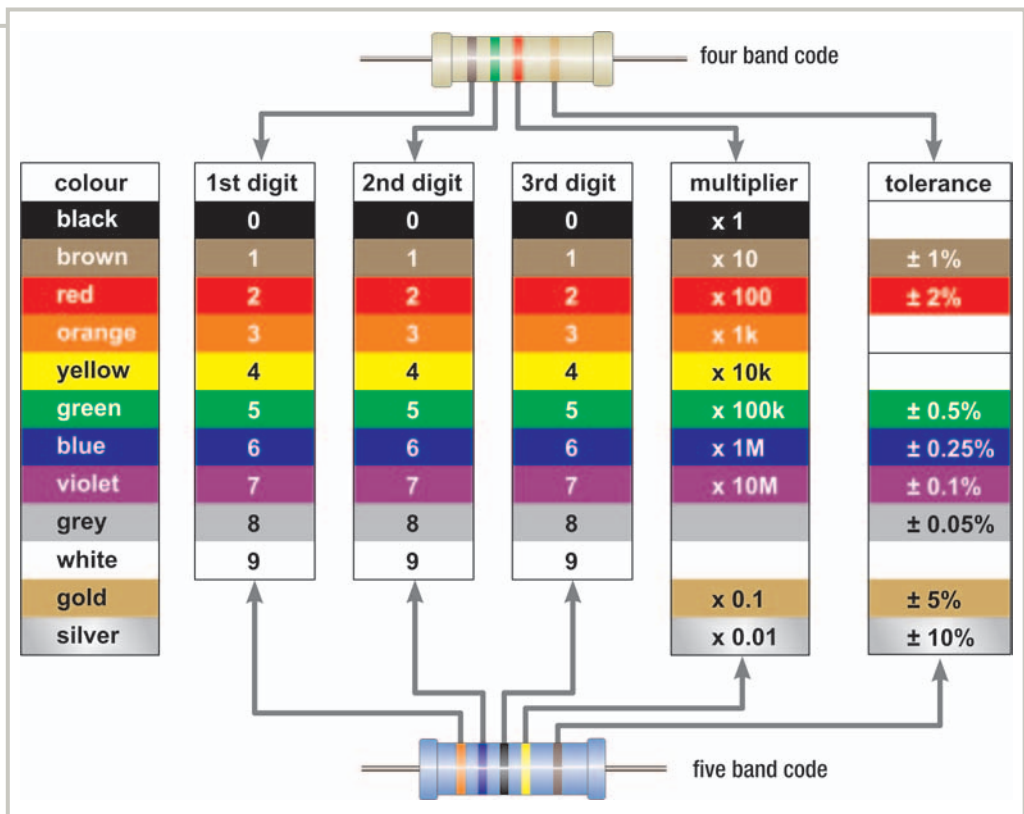
The examples in Figure 5.18 show how to use the colour code. The first two bands (or three for precision resistors) are given a number, the next band is the multiplier and the last (on the right) is the tolerance band.

Here are a few points that will help you use the colour code. Practice soon makes the process easy. If in doubt (as is often the case with precision resistors), check the resistance with an ohmmeter!

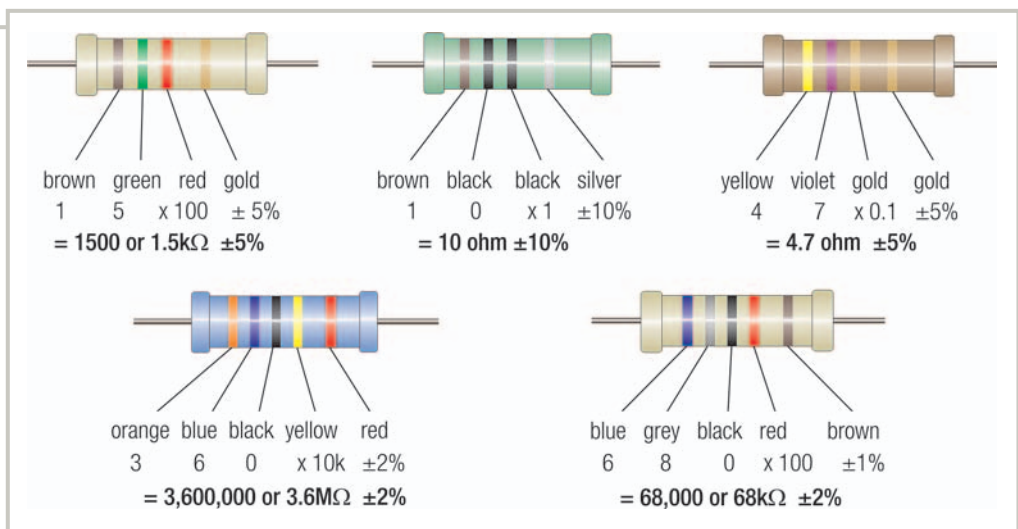
- The tolerance band is spaced further apart from the other bands.
- The digit bands are generally closer to one end of the resistor. They're read from left to right and the band nearest the end of the resistor is the first digit.
- The first band can never be black, gold or silver.
- Resistors with four bands have either a gold or, less often, a silver tolerance band.

FIGURE 5.17

Resistor colour code

**FIGURE 5.18**

Using the resistor colour code



- Resistors with five bands usually have either a red or brown tolerance band.
- The multiplier band is unlikely to be violet (7), grey (8) or white (9).
- You might be able to remember the colour sequence with this mnemonic:

Black **B**erries **R**oam **O**ver **Y**our **G**arden **B**ut **V**iolets **G**row **W**ild

An unusual colour code is the one for the *zero-ohm* resistor. This resistor, shown in Figure 5.19, has a single black band, which represents the number 0. Instead of carbon, the resistance element is a piece of copper wire. The main reason for a zero-ohm resistor is so an automatic 'pick and place' machine can handle them and insert what amounts to a link in a printed circuit board.

Otherwise, wire links need to be put in by hand, which increases costs and slows the process. (The surface mount resistors marked 0 in Figure 5.6 are also zero ohm resistors.)

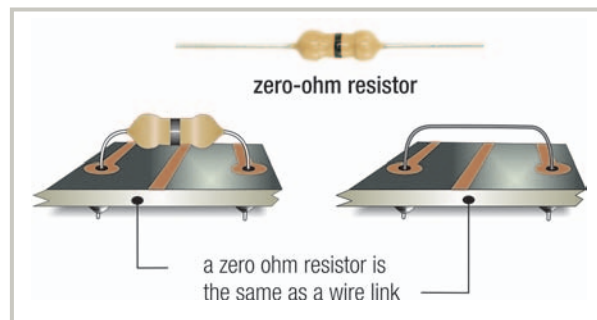


FIGURE 5.19

The zero-ohm resistor is equivalent to a piece of wire

Preferred resistor values

Resistor values range from 0.01 ohms to over 80 million ohms. It's not economical for resistor manufacturers to make every possible value – this would require over 80 million different resistors.

Instead, a standard set of values, called the *preferred values*, is made. These were chosen when resistors typically had a tolerance of 10 per cent. The values were determined on the basis that because a resistor might have an actual resistance 10 per cent higher or lower than its marked value, one resistor therefore covered a range of values. By selecting suitable base values, a total of 12 different resistors could cover a decade range. A decade range is, for example, all the numbers between 1 and 9, or between 10 and 99, 100 to 999 and so on.

The preferred values in the E12 range (so called because there are 12 values per decade) are in Table 5.4. These values are those generally stocked by most electronic parts suppliers. In Table 5.4:

- 10 decades are covered (from $\times 0.01$ to 10 000 000 or 10 M)
- there are 12 resistor values available in a decade (the preferred values)
- the 12 preferred values are 1, 1.2, 1.5, 1.8, 2.2, 2.7, 3.3, 3.9, 4.7, 5.6, 6.8 and 8.2
- there are 120 possible resistor values, ranging from 0.01Ω to $82 \text{ M}\Omega$

E12 PREFERRED RESISTOR VALUES

TABLE 5.4

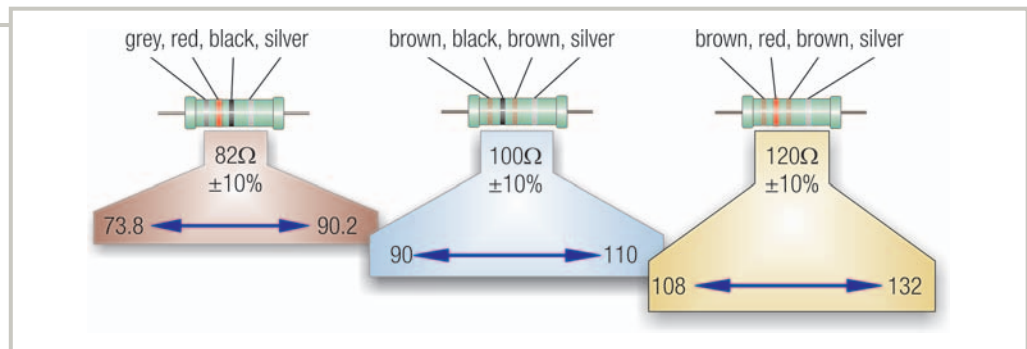
Decade	Range of preferred values											
$\times 0.01$	0.01	0.012	0.015	0.018	0.02	0.027	0.033	0.039	0.047	0.056	0.068	0.082
$\times 0.1$	0.1	0.12	0.15	0.18	0.22	0.27	0.33	0.39	0.47	0.56	0.68	0.82
$\times 1$	1	1.2	1.5	1.8	2.2	2.7	3.3	3.9	4.7	5.6	6.8	8.2
$\times 10$	10	12	15	18	22	27	33	39	47	56	68	82
$\times 100$	100	120	150	180	220	270	330	390	470	560	680	820
$\times 1 \text{ k}$	1 k	1.2 k	1.5 k	1.8 k	2.2 k	2.7 k	3.3 k	3.9 k	4.7 k	5.6 k	6.8 k	8.2 k
$\times 10 \text{ k}$	10 k	12 k	15 k	18 k	22 k	27 k	33 k	39 k	47 k	56 k	68 k	82 k
$\times 100 \text{ k}$	100 k	120 k	150 k	180 k	220 k	270 k	330 k	390 k	470 k	560 k	680 k	820 k
$\times 1 \text{ M}$	1 M	1.2 M	1.5 M	1.8 M	2.2 M	2.7 M	3.3 M	3.9 M	4.7 M	5.6 M	6.8 M	8.2 M
$\times 10 \text{ M}$	10 M	12 M	15 M	18 M	22 M	27 M	33 M	39 M	47 M	56 M	68 M	82 M

- to calculate any resistor value, multiply the preferred value by the decade
- the lowest resistor value is $0.01\ \Omega$ and the highest is $82\ \text{M}\Omega$.

For example, as shown in Figure 5.20, you can get all resistance values from 73.8 to $132\ \Omega$ by selecting them from a batch of 82 , 100 or $120\ \Omega \pm 10\ \text{per cent}$ resistors.

FIGURE 5.20

These three 10 per cent resistors give all values from $73.8\ \text{ohm}$ to $132\ \text{ohm}$



The E12 range of preferred resistors assumes a tolerance in the resistor value of 10 per cent; however, most resistors now have a 5 per cent tolerance. Therefore, if 5 per cent resistors are used, the E12 preferred values don't cover all possible resistance values.

While the E12 preferred resistance values suit most applications, a wider range of resistors is made, called the *E24 preferred value range*. This range is based on 5 per cent tolerance resistors and has 24 values in each decade. The values are shown below. Notice how this range includes the E12 range (in *italic*) with 12 extra values (underneath them in **bold**), giving a total of 24 resistance values.

TABLE 5.5

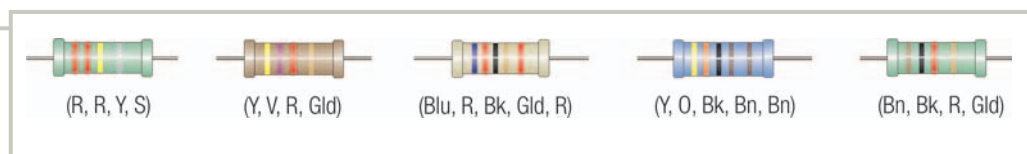
THE E24 RANGE OF RESISTORS												
<i>E12:</i>	<i>1</i>	<i>1.2</i>	<i>1.5</i>	<i>1.8</i>	<i>2.2</i>	<i>2.7</i>	<i>3.3</i>	<i>3.9</i>	<i>4.7</i>	<i>5.6</i>	<i>6.8</i>	<i>8.2</i>
	1.1	1.3	1.6	2.0	2.4	3	3.6	4.3	5.1	6.2	7.5	9.1

EXERCISE 5.2

- 1 Use the resistor colour code to determine the values of the resistors in Figure 5.21 below.

FIGURE 5.21

Practice exercise for using the resistor colour code



5.4 Measuring resistance

Resistance is measured with an *ohmmeter*, which can be either an analog or a digital instrument. These are explored further in Chapter 9. Most ohmmeters are part of a multimeter, in which one or two switches select the range and function (volts, current or resistance). There are two important things to be aware of when measuring resistance with an ohmmeter:

- 1 If you are measuring resistance in a circuit, make sure the circuit is not connected to any type of power source, in particular $230\ \text{V}$ mains. If there is a voltage in the circuit, it can cause damage to the meter. In extreme cases, the meter can explode, catch fire and possibly cause injury.

- 2 While measuring the resistance of a component, do not touch both probes at the same time, as the reading will be affected by your body resistance.

To measure resistance with a multimeter:

- 1 Select the resistance (OHMS or Ω) function, normally by a rotary selector switch.
- 2 For an analog ohmmeter, zero the pointer for both no deflection and full deflection, as shown in Figure 5.22. Make the full deflection adjustment each time you change the range.
- 3 Select a range. Many digital multimeters are auto-ranging, although, like the analog multimeter, some require a range to be selected. For an analog meter the best range will give a reading between a quarter and three-quarters of full scale. For a digital meter, select a range closest to the value of the resistor. For example, for a 1 k Ω resistor, a scale of 2 k Ω is suitable.



A live circuit and an ohmmeter are a dangerous combination

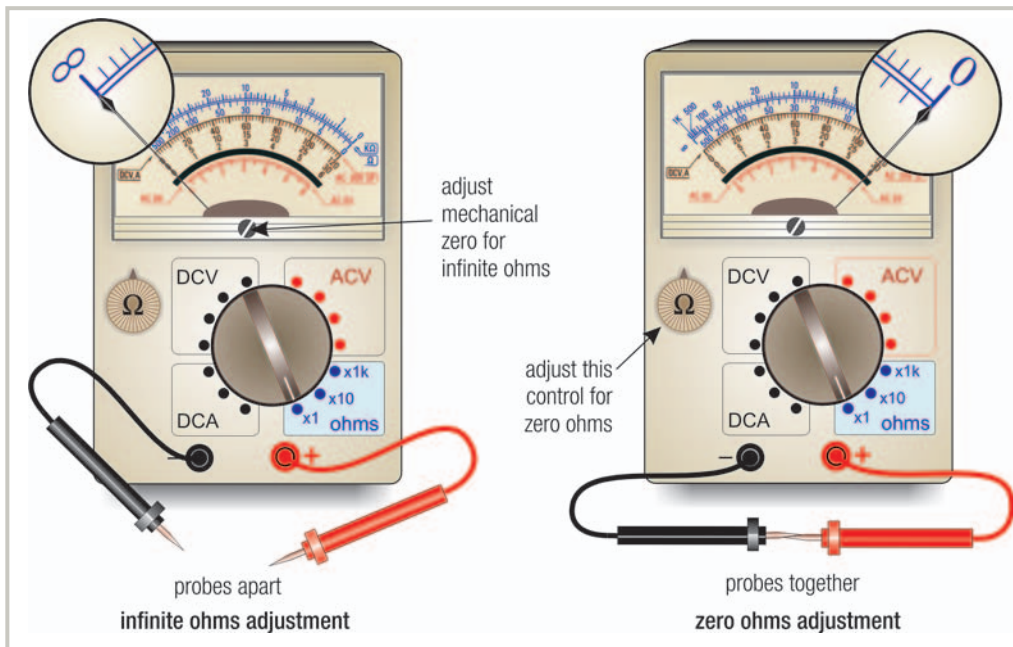


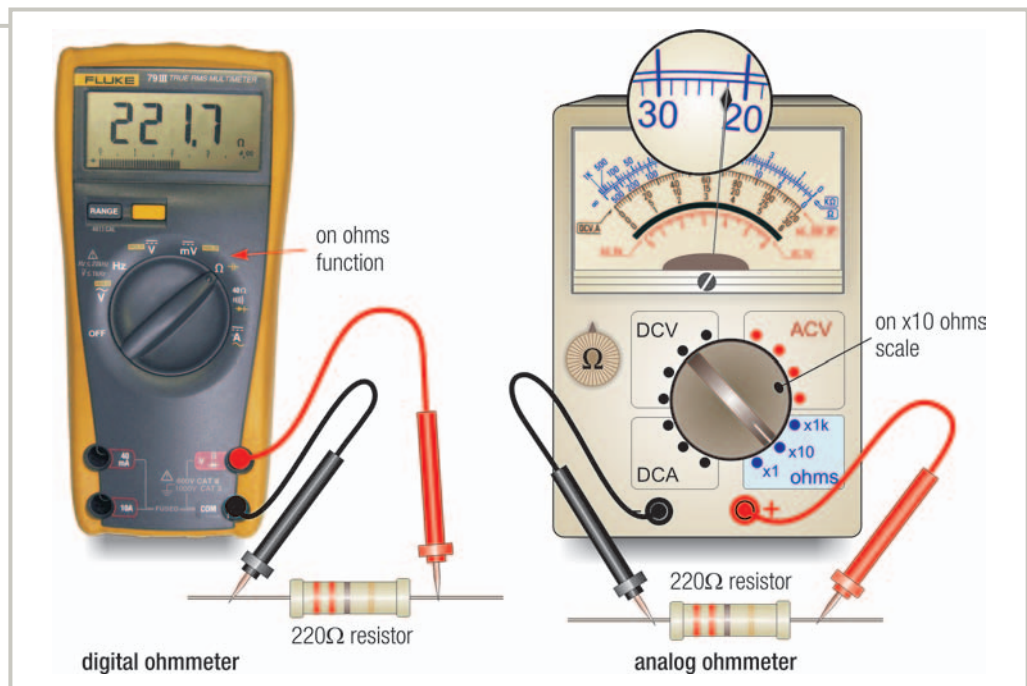
FIGURE 5.22

Adjust the mechanical zero if the pointer is not at infinite ohms when the probes are apart. With the probes touching, adjust the meter's 'zero ohm' control for 0 Ω .

- 4 Connect the ohmmeter probes across the resistor, or to the points in a circuit, and if necessary select a more suitable range. For an analog meter, if the pointer only moves a little, select a higher resistance range. If the pointer is close to full scale, choose a lower range.
- 5 Read the resistance value. On analog meters, the resistance reading is the scale reading multiplied by the range. For example, a reading of 50 on the scale for a range of OHMS $\times$ 100 means the resistance is 50×100 , or 5000 Ω . See Figure 5.23.

FIGURE 5.23

Analog and digital multimeters measuring the resistance of a 220 Ω resistor



CHAPTER SUMMARY

- Resistance of a conductor can be calculated with the equation $R = \frac{\rho l}{A}$, where ρ is resistivity in ohm-metre, l is length in metres and A is cross-sectional area in square metres.
- Conductors such as copper and aluminium have a positive temperature coefficient of 0.004, which means the resistance increases by 0.4 per cent per 1 $^{\circ}\text{C}$.
- Resistors are electrical components that range from large, metal grid units to surface mount devices. They are rated by their power and resistance values. Low power resistors have their resistance identified with a colour code. The resistor colour code is based on the colour spectrum, but starting at black and ending at white.
- The tolerance of a resistor indicates the spread of resistance values a resistor might actually have.
- Low power resistors are generally manufactured in either the E12 or E24 ranges.
- Components that change resistance due to an external influence include the variable resistor (mechanical force), LDR (light), thermistor (temperature) and the varistor (voltage).
- When using an ohmmeter, ensure the circuit is 'dead' and only touch one metal probe at a time.

SELF-CHECK

Answer these questions then check your answers at the back of the book.

- 1 A 100 m long copper cable is allowed to have a resistance no more than one ohm. What is the minimum cross-sectional area of a suitable cable?
- 2 A relay coil made of copper has a resistance of 200 Ω at 20 $^{\circ}\text{C}$. What is its approximate resistance at 100 $^{\circ}\text{C}$?
- 3 Which has the highest resistance: a cable with a cross-sectional area of 4 mm^2 or a cable of the same length but with a diameter of 2 mm ?

- 4 For Question 3, what is the difference between the resistance values of the cables if they are both 1 km long?
- 5 What is meant by the term 'NTC' in regard to resistance?
- 6 Determine the maximum current that can flow in a $100\ \Omega$ resistor rated at 10 W.
- 12 Give two typical applications of a thermistor.
- 13 What is the difference between a surge protector and a surge arrestor?
- 14 What is the colour code of a $5.6\ \Omega$ resistor with a tolerance of 5 per cent?
- 15 Determine the resistance and tolerance of each resistor in Figure 5.24 below.

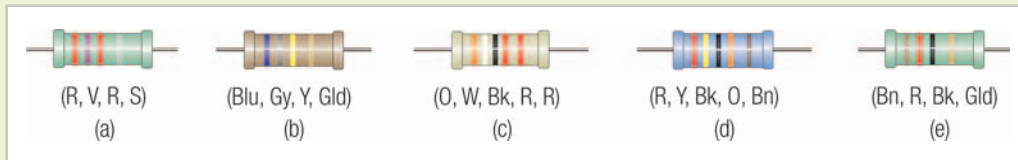


FIGURE 5.24

- 7 What type of construction would the resistor in Question 6 be likely to have?
- 8 Is the temperature coefficient of resistance of most semiconductors positive or negative?
- 9 A 100 k ohm variable resistor when set halfway measures 2 k ohm between the centre terminal and one end terminal. Is the variable resistor linear or non-linear?
- 10 A grid resistor has a power rating of 6 kW and a rated current of 100 A. What is its resistance?
- 11 What happens to the resistance of an LDR as the light intensity increases?
- 16 What is the maximum allowable resistance of a $680\ \Omega$, 10 per cent resistor?
- 17 What is the resistance and tolerance of a five band resistor with all bands coloured red?
- 18 What is meant by the E12 'preferred' range of resistor values?
- 19 Why is there also an E24 preferred range of resistor values?
- 20 When measuring resistance, why shouldn't you touch the probes of an ohmmeter?

ONLINE RESOURCES

For more information about the contents of this chapter, try Googling these terms:

- resistor (recommended websites include Wikipedia and www.allaboutcircuits.com)
- grid resistor (Australian website www.resistor-repairs.com.au/index.htm gives photos and description of units used in the mining industry)
- resistor colour code (some websites give colour code mnemonics we dare not print).